

Computer Vision in Mining

The Big Picture

Where vision AI helps across the mining value chain.



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Why it matters



Safety

Helps keep people and assets safe.



Productivity

Increases efficiency and reduces downtime.



Quality & Recovery

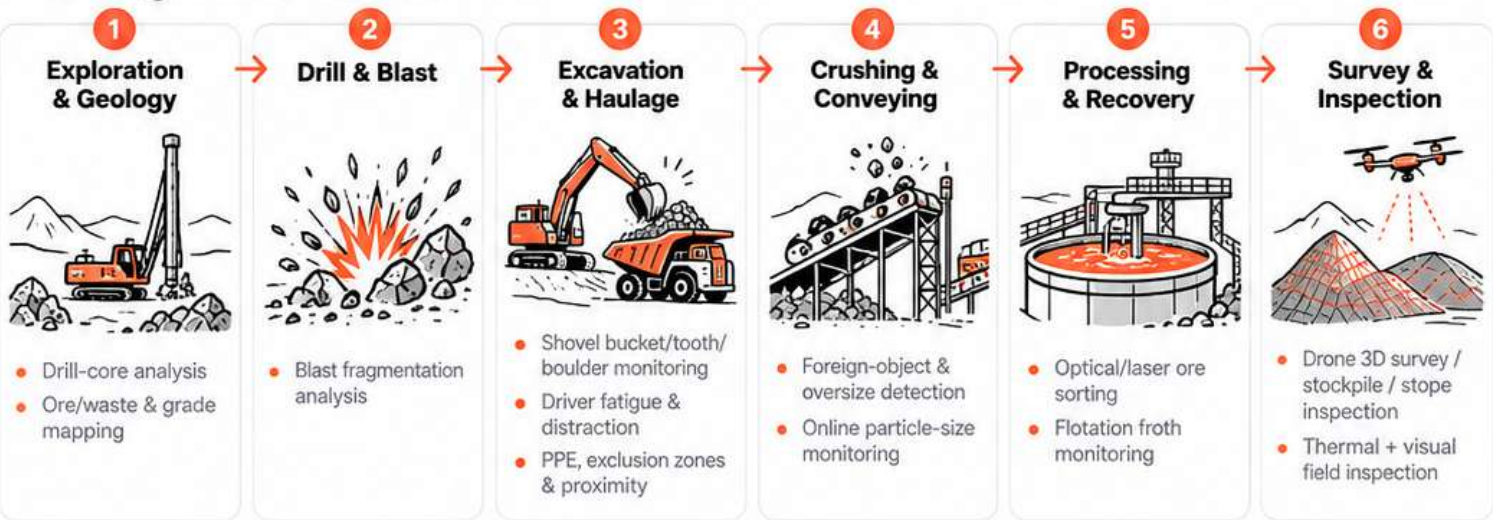
Improves grade, recovery and consistency.



Inspection & Monitoring

Enables continuous, real-time visibility.

Mining value chain



What CV does



Detect

Find and classify objects and features.



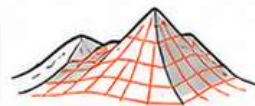
Measure

Quantify size, shape distance and volume.



Track

Follow movement and behaviour over time.



Reconstruct

Build 3D models and digital twins.



Inspect

Identify defects, changes and anomalies.



Computer vision in mining is not one tool – it spans safety, geology, process control and operational intelligence.



Use Cases 1-4

Rock, Ore and Fragmentation

Early-stage and mining-face applications.

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1 Drill-core analysis



	Input	Core tray photos.
	CV task	Detect fractures, veins, discing and lithology features.
	Output	Structured geological and geotechnical data.
	Why it matters	Faster logging and scalable review of historical core.
	Real example	Datarock.

2 Ore / waste & grade mapping



	Input	Face or stockpile imagery plus hyperspectral or spectral sensing.
	CV task	Classify material zones and grade proxies.
	Output	Ore/waste map and selective mining guidance.
	Why it matters	Reduce dilution and improve cut-off decisions.
	Real example	OreSense trials.

3 Blast fragmentation analysis



	Input	Muckpile, shovel or conveyor images.
	CV task	Segment fragments and estimate PSD, P20, P50 and P80.
	Output	Fragmentation metrics.
	Why it matters	Better digging, crusher feed and mill performance.
	Real example	Orica FRAGTrack / Newmont Akyem.

4 Shovel bucket / tooth / boulder monitoring



	Input	Shovel camera.
	CV task	Detect missing teeth, boulders and loading conditions.
	Output	Alerts and productivity insights.
	Why it matters	Protect crushers and improve loading efficiency.
	Real example	Weir ShovelMetrics.



These use cases turn rock images into geological insight, fragmentation control and loading intelligence.



Use Cases 5-8

Safety and Material Handling

Real-time monitoring around people, equipment and material flow.



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5 Driver fatigue & distraction



	Input	In-cab camera.
	CV task	Analyse face, eyes and head behaviour over time.
	Output	Real-time fatigue or distraction event.
	Why it matters	Safer haulage and fewer incidents.
	Real example	Hexagon / MMG Rosebery.

6 PPE, exclusion zones & proximity



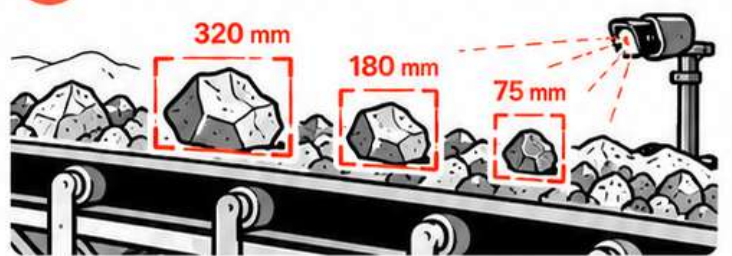
	Input	CCTV or mobile cameras.
	CV task	Detect people, PPE and vehicles, then track entries into danger zones.
	Output	Violation or near-miss alerts.
	Why it matters	Worker safety and safer interactions around equipment.
	Real example	—

7 Conveyor foreign-object & oversize detection



	Input	Fixed belt camera.
	CV task	Detect metal, wood, tools, oversize rocks or material anomalies.
	Output	Early warning before crusher damage or downtime.
	Why it matters	Asset protection and less unplanned stoppage.
	Real example	BHP OFOD; Boliden / Norrin.

8 Online particle-size monitoring



	Input	Belt or chute camera.
	CV task	Segment particles and estimate size distribution continuously.
	Output	PSD trend and coarse/fine changes.
	Why it matters	Process stability and better downstream control.
	Real example	Metso RockSense.

This block focuses on **protecting people**, **protecting assets** and **stabilising material handling**.

Use Cases 9-12

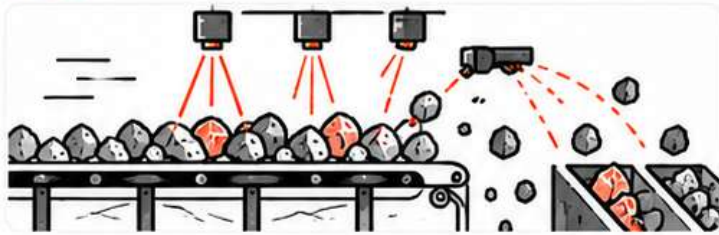
Processing, Survey and Inspection

Advanced CV for mineral processing, 3D mapping and field inspection.

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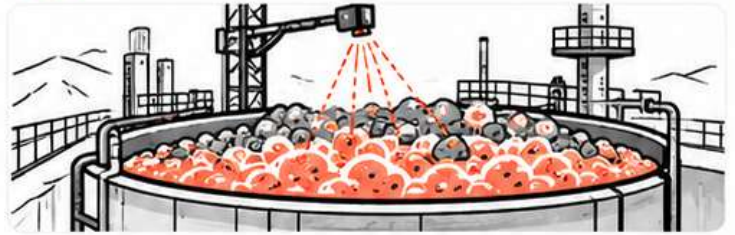


9 Optical / laser ore sorting



	Input	Rocks on a sorter with optical, laser or spectral sensing.
	CV task	Classify each particle as keep or reject.
	Output	Timed air-jet separation and pre-concentration.
	Why it matters	Remove waste early and reduce downstream processing load.
	Real example	TOMRA Mt Cattlin.

10 Flotation froth monitoring



	Input	Video above a flotation cell.
	CV task	Analyse bubble size, velocity, stability and texture.
	Output	Process metrics for control.
	Why it matters	Improve recovery, consistency and operator visibility.
	Real example	Metso FrothSense+.

11 Drone 3D survey / stockpile / stope inspection



	Input	Overlapping drone images.
	CV task	Photogrammetry and 3D reconstruction.
	Output	Point cloud, mesh, volume and overbreak insights.
	Why it matters	Safer survey work and faster inventory or geotechnical review.
	Real example	Barrick Golden Sunlight.

12 Thermal + visual field inspection



	Input	RGB and thermal drone images.
	CV task	Classify wet, dry or abnormal zones and locate thermal anomalies.
	Output	Map of problem locations.
	Why it matters	Large-area inspection and targeted maintenance.
	Real example	SQM + FlytBase.



These use cases show that mining CV goes beyond CCTV — it also includes **process vision, photogrammetry and thermal inspection.**



How the Technology Works

Core CV methods + real mining deployments

From detection to 3D reconstruction, different mining problems need different vision methods.



Core CV methods

1. Detect



Find people, tools, boulders and objects.

2. Segment & Measure



Separate fragments or particles and quantify size.

3. Track



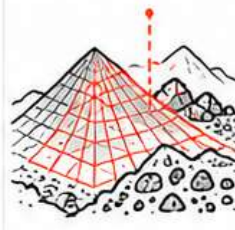
Follow people, vehicles and movement over time.

4. Temporal Vision



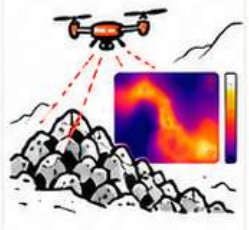
Analyse fatigue behaviour or froth dynamics.

5. Reconstruct in 3D



Build point clouds, meshes and volumes from images.

6. Thermal & Spectral Sensing



Use heat patterns or wavelength signatures when RGB is not enough.

Public mining deployments

Datarock

Core discing and fracture analysis



Orica FRAGTrack / Newmont Akyem

Fragmentation



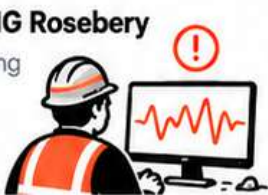
Weir ShovelMetrics

Shovel teeth, boulders, loading insight



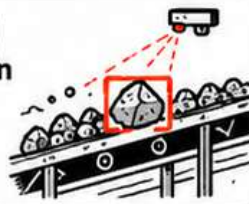
Hexagon / MMG Rosebery

Fatigue monitoring



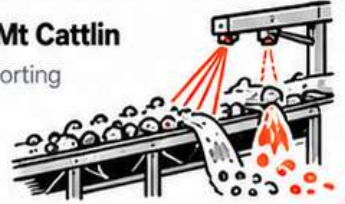
BHP OFOD and Boliden / Norrín

Conveyor hazard detection



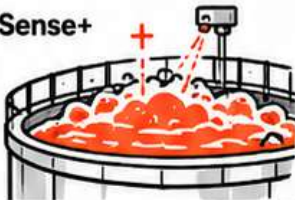
TOMRA Mt Cattlin

Laser ore sorting



Metso FrothSense+

Flotation froth monitoring



Barrick Golden Sunlight

Drone stope inspection



SQM / FlytBase

RGB + thermal inspection



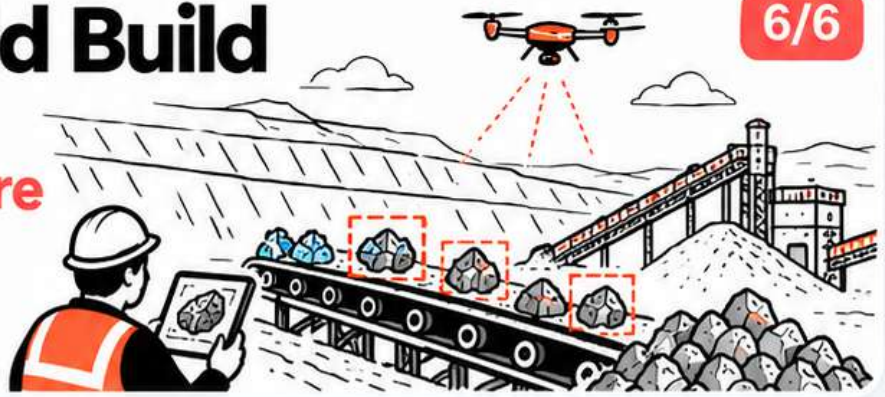
The right sensor and CV method depend on the physical problem, **not just the model name.**



What We Should Build

Top projects + production architecture

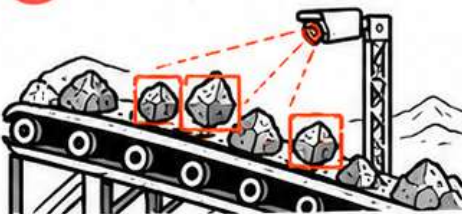
Recommended starting points for a practical mining-CV roadmap.



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Top 3 software-first opportunities

1 Conveyor Material Vision



- Detect foreign objects and oversize material; later add particle-size monitoring.

★ **Why it is strong:** Fixed camera, clear ROI, high practical value.

2 Blast Fragmentation Intelligence



- Estimate PSD and P80 from calibrated imagery.

★ **Why it is strong:** Highly mining-specific and useful for blast feedback.

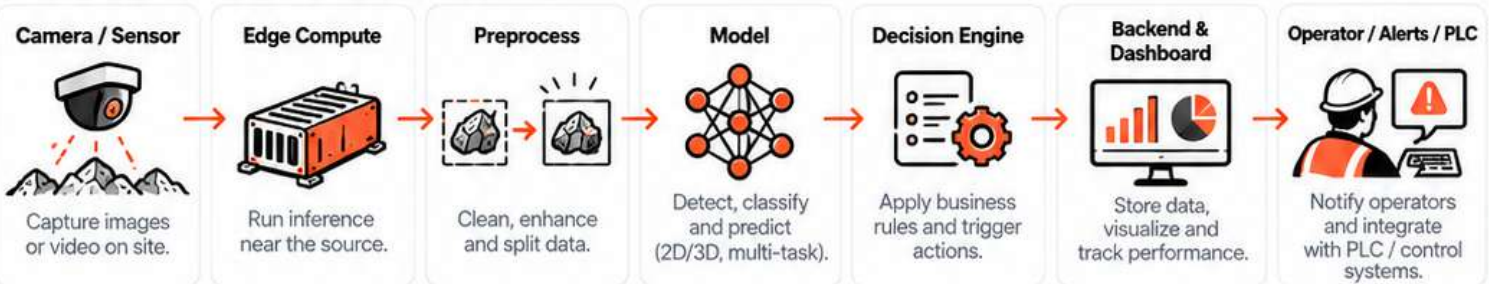
3 Drill-Core Vision



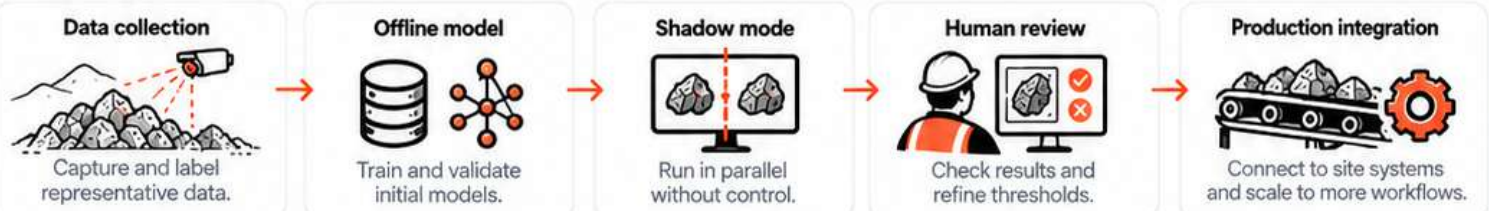
- Convert core photographs into fracture, discing and geological insight.

★ **Why it is strong:** Leverages existing image archives.

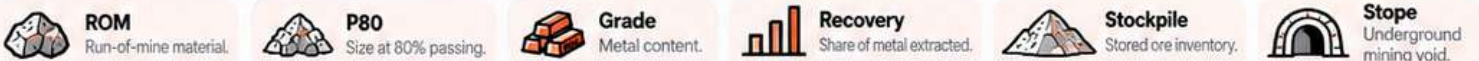
Production architecture



Deployment path



Mini glossary



**Start with one workflow, validate on site,
then scale to a broader mining-vision platform.**